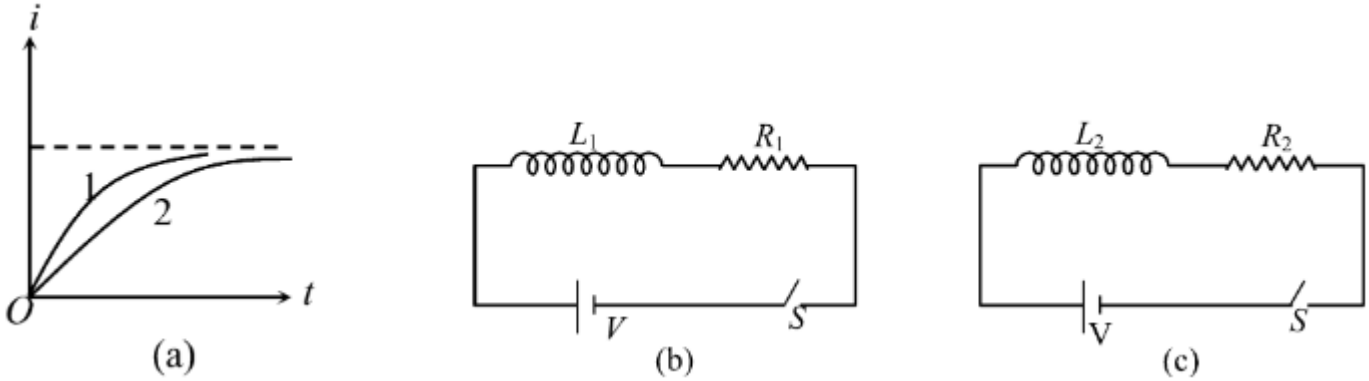
Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Current grows into L-R circuits (b) and (c) as shown in figure (a). Let L_1, L_2, R_1 and R_2 be the corresponding values of inductance and resistance in two circuits, then



(A) $R_1 > R_2$

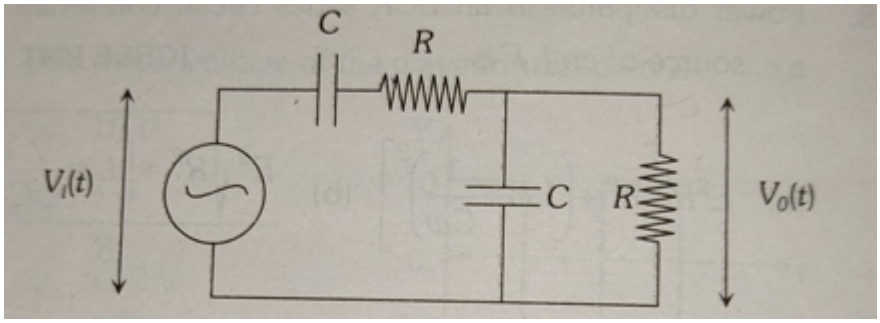
(B) $R_1 < R_2$

(C) $L_1 > L_2$

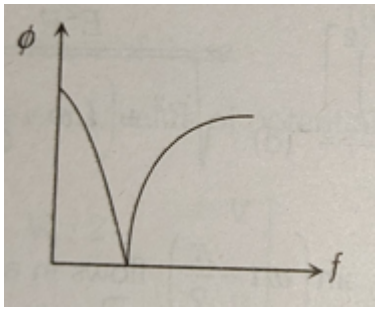
(D) $L_1 < L_2$

Q.2

If the input voltage V_i to the circuit below is given by $V_i(t) = A\cos(2\pi ft)$ the output voltage is given by $V_0(t) = B\cos(2\pi ft + \phi)$. Which one of the following four graphs best depict the variation of ϕ vs. f

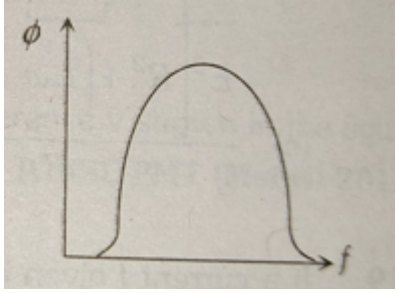


(A)

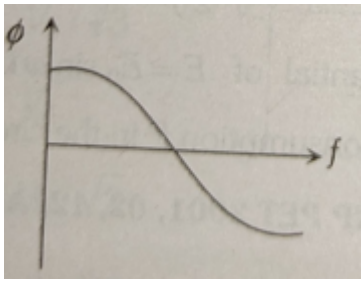


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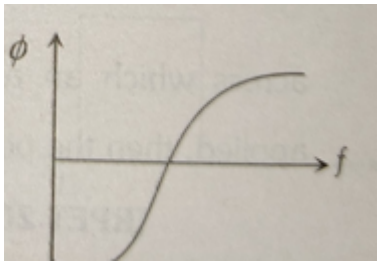
(B)



(C)

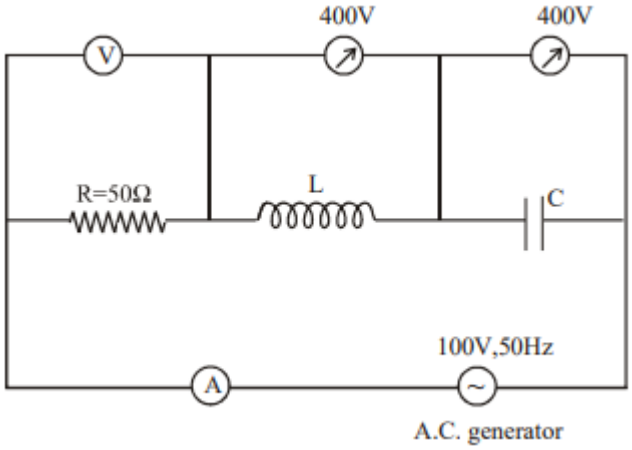


(D)



Q.3

In the LCR series circuit, the voltmeter and ammeter readings are -



(A) $E = 100$ volts, $I = 2$ amp

(B) $E = 100$ volts, $I = 5$ amp

(C) $\mathrm{E} = 300$ volts, $\mathrm{I} = 2\mathrm{amp}$

(D) $E = 300$ volts, $I = 5$ amp

Q.4

If the total charge stored in the LC circuit is Q_0, then for $t \geq 0$

(A) The charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} + \frac{t}{\sqrt{LC}}\right)$

(B) The charge on the capacitor is $Q = Q_0 \cos\left(\frac{\pi}{2} - \frac{t}{\sqrt{LC}}\right)$

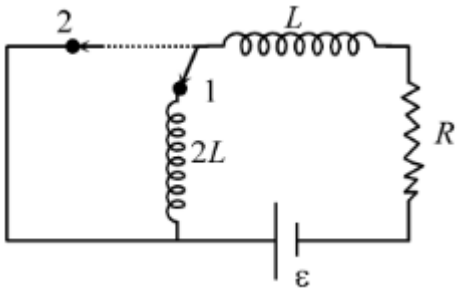
(C) The charge on the capacitor is $Q = -LC\frac{d^2Q}{dt^2}$

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(D) The charge on the capacitor is $Q = \frac{1}{\sqrt{LC}} \frac{d^2Q}{dt^2}$

Q.5

In the circuit shown switch is connected to 1 for a very long time. At a particular instant $t = 0$, switch is shifted to 2, the current in the circuit after a time gap of $\frac{L}{R}$ is



(A) $\frac{\varepsilon}{2\mathrm{R}e}$

(B) $\frac{3\varepsilon}{\mathrm{R}}$

(C) $\frac{3\varepsilon}{\mathrm{R}e}$

(D) $\frac{\varepsilon}{R}$

Q.6

In an $\mathrm{L}-\mathrm{R}$ circuit, the value of L is $(0.4/\pi)$ henry and the value of R is $30\mathrm{ohm}$. If in the circuit, an alternating emf of 200 volt at 50 cycles per second is connected, the impedance of the circuit and current will be :

(A) $11.4\mathrm{ohm}, 17.5$ ampere

(B) $30.7\mathrm{ohm}, 6.5$ ampere

(C) $40.4\mathrm{ohm}, 5$ ampere

(D) $50\mathrm{ohm}, 4$ ampere.

Q.7

A R-L circuit draws a power of $440\ \mathrm{W}$ from a source of $220\ \mathrm{V}, 50\ \mathrm{Hz}$. The power factor of the circuit is 0.5. To make the power factor of the circuit as 1.0, what capacitance will have to be connected with it ?

(A) $4.68 \times 10^{-5}\ \mathrm{F}$

(B) $6.68 \times 10^{-5}\ \mathrm{F}$

(C) $6.68 \times 10^{-3}\ \mathrm{F}$

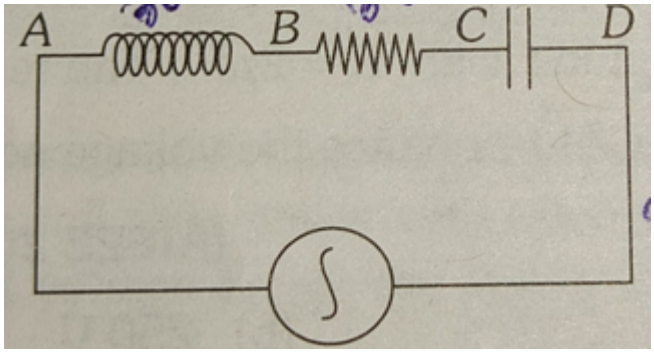
(D) $7.68 \times 10^{-5}\ \mathrm{F}$

Q.8

An ac voltmeter connected between points A and B in the circuit below reads $36\ \mathrm{V}$. If it is connected between A and C, the reading is $39\ \mathrm{V}$. The reading when it is connected between B and D is $25\ \mathrm{V}$. What will the voltmeter reading when it is connected between A and D (Assume that the voltmeter reads true rms voltage values and that the

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source generated a pure ac)



(A) $\sqrt{481}$ V

(B) 31 V

(C) 61 V

(D) $\sqrt{3361}$ V

Q.9

A resistance R and inductance L and a capacitor C all are connected in series with an AC supply. The resistance of R is 16 ohm and for a given frequency, the inductive reactance of L is 24 ohm and capacitive reactance of C is 12ohm. If the current in the circuit is 5 amp.. Choose the correct option

(A) Potential difference across R is 80 V

(B) Potential difference across L is 120 V

(C) Potential difference across C is 60 V

(D) Potential difference across C is 80 V

Q.10

The instantaneous voltages at three terminals marked X, Y and Z are given by $V_X = V_0 \sin \omega t$
$V_Y = V_0 \sin\left(\omega t + \frac{2\pi}{3}\right)$ and $V_Z = V_0 \sin\left(\omega t + \frac{4\pi}{3}\right)$
An ideal voltmeter is configured to read rms value of the potential difference between its terminals. It is connected between points X and Y and then between Y and Z. The reading(s) of the voltmeter will be

(A) $V_{XY}^{rms} = V_0\sqrt{\frac{3}{2}}$

(B) $V_{YZ}^{rms} = V_0\sqrt{\frac{1}{2}}$

(C) Independent of the choice of the two terminals

(D) $V_{XY}^{rms} = V_0$

Q.11

The value of current in two series LCR circuits at resonance is same when connected across a sinusoidal voltage source. Then choose the incorrect options -

(A) both circuits must be having same value of capacitance and inductor

(B) in both circuits ratio of L and C will be same

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(C) for both the circuits X_L/X_C must be same at that frequency

(D) both circuits must have same impedance at all frequencies.

Q.12

Passage

When an alternating voltage of $220\ \mathrm{V}$ is applied across a device X, a current of 0.5 A flows through the circuit and is in phase with the applied voltage. When the same voltage is applied across another device Y, the same current again flows through the circuit but it leads the applied voltage by $\pi/2$ radians.

Question

Name the devices X

(A) resistor

(B) capacitor

(C) inductor

(D) None

Q.13

Passage

When an alternating voltage of $220\ \mathrm{V}$ is applied across a device X, a current of 0.5 A flows through the circuit and is in phase with the applied voltage. When the same voltage is applied across another device Y, the same current again flows through the circuit but it leads the applied voltage by $\pi/2$ radians.

Question

Name the devices Y.

(A) resistor

(B) capacitor.

(C) inductor

(D) None

Q.14

Passage

When an alternating voltage of $220\ \mathrm{V}$ is applied across a device X, a current of 0.5 A flows through the circuit and is in phase with the applied voltage. When the same voltage is applied across another device Y, the same current again flows through the circuit but it leads the applied voltage by $\pi/2$ radians.

Question

Calculate the current flowing in the circuit when same voltage is applied across the series combination of X and Y.

(A) $0.35\ \mathrm{A}$

(B) $0.25\ \mathrm{A}$

(C) $0.05\ \mathrm{A}$

(D) $0.45\ \mathrm{A}$

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Q.15

Passage

For a series LCR circuit $\mathrm{I}=100 \sin (100 \pi \mathrm{t}-\pi / 3) \mathrm{mA}$ and $\mathrm{V}=100 \sin (100 \pi \mathrm{t})$ volt, then

Question

Calculate resistance of circuit.

(A) 1000Ω

(B) $500 \sqrt{3} \Omega$

(C) 500Ω

(D) None of these

Q.16

Passage

For a series LCR circuit $\mathrm{I}=100 \sin (100 \pi \mathrm{t}-\pi / 3) \mathrm{mA}$ and $\mathrm{V}=100 \sin (100 \pi \mathrm{t})$ volt, then

Question

Calculate reactance of circuit.

(A) 1000Ω

(B) $500 \sqrt{3} \Omega$

(C) 500Ω

(D) None of these

Q.17

Passage

For a series LCR circuit $\mathrm{I}=100 \sin (100 \pi \mathrm{t}-\pi / 3) \mathrm{mA}$ and $\mathrm{V}=100 \sin (100 \pi \mathrm{t})$ volt, then

Question

Find average power loss.

(A) $2.5 \mathrm{~W}$

(B) $1.5 \mathrm{~W}$

(C) $3.5 \mathrm{~W}$

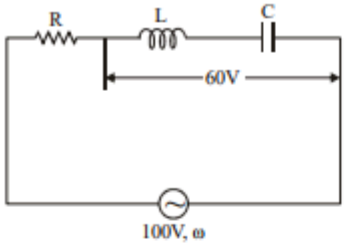
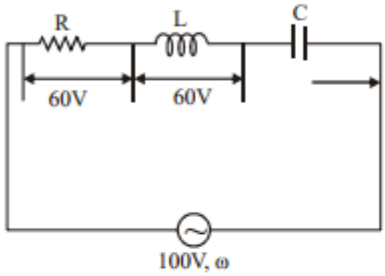
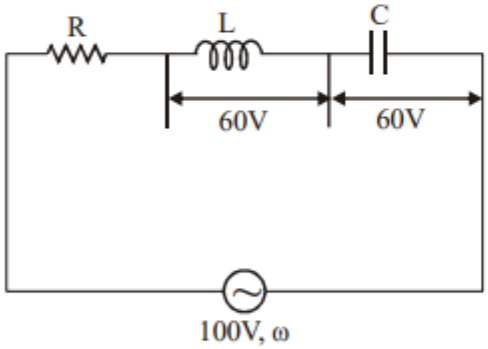
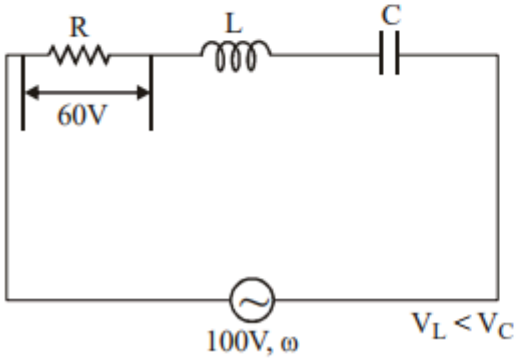
(D) $4.5 \mathrm{~W}$

Q.18

Column I gives series RLC-A.C. circuits $R, L, C>0$. Voltages given are rms values. Match them with

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conclusions given in column II. Only if they can be certainly drawn.

Column I	Column II
(A) 	(p) Currents leads voltage
(B) 	(q) Current does not lead voltage
(C) 	(r) ω < resonant frequency
(D) 	(s) Power factor > 0.5
	(t) Impedance $\geq \frac{5}{4}$ R

(A) $(A)-s,t(B)-p,r,s,t(C)-q,s(D)-p,r,s,t$

(B) $(A)-p,t(B)-p,r,t(C)-q,s(D)-p,r,s$

(C) $(A)-s,t(B)-r,s,t(C)-q \quad (D)-p,r$

(D) $(A)-p,s,t(B)-p,r,s,t(C)-q,s(D)-t$

Q.19

At a hydroelectric power plant, the water pressure head is at a height of $300\ \mathrm{m}$ and the water flow available is $100\ \mathrm{m}^3/\mathrm{s}$. If the turbine-generator efficiency is 60%, estimate the electric power available from the plant $\left(g = 9.8\ \mathrm{m/s^2}\right)$

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Q.20

An a.c. source has an internal resistance of $10^4\Omega$. Calculate the turns ratio of a transformer so as to match the source to a load of resistance 100Ω. Express your answer in powers of 10^{-1}. Assume the transformer to be ideal.

Answers & Solutions

Q.1 Answer:

$L_1 < L_2$

Solution:

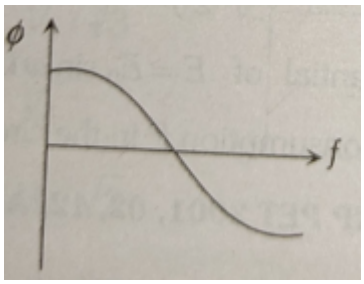
Steady state current for both the circuits is same.

Therefore $\frac{V}{R_1} = \frac{V}{R_2}$ **or** $R_1 = R_2$

Further $\tau_{L_1} < \tau_{L_2}$ $(\tau_L =$ **time constant)**

$\therefore \quad \frac{L_1}{R_1} < \frac{L_2}{R_2}$ or $L_1 < L_2 \quad (R_1 = R_2)$

Q.2 Answer:



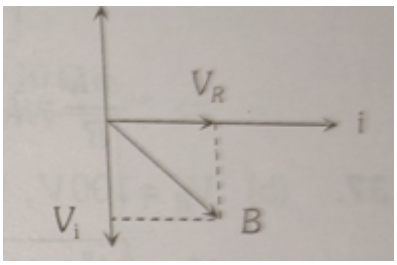
Solution:

Resultant of V_i, V_R and B give V_1 and angle between

V_i **and** B **is** ϕ

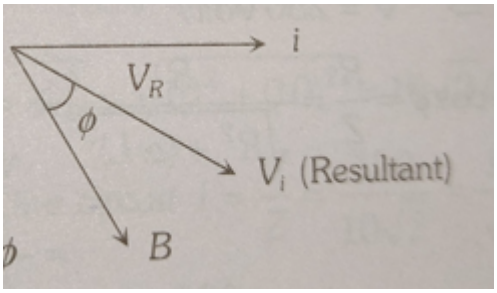
When f **is very high**

$$X_C = \frac{1}{\omega c} \to 0, \text{ then } V_i = 0$$



Resultant of V_i, V_R **and** B **between** BV_R

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B lag behind ϕ

At higher frequency ϕ become -ve.

Q.3 Answer:

$E = 100$ volts, $I = 2$ amp

Solution:

V_L and V_C will be in opposite phase so they will cancel each other being equal in magnitude.

$\therefore$ **Resultant potential difference = applied p.d. = 100 volts**

Hence, $Z = R$

$I_{rms} = \frac{V_{rms}}{Z} = \frac{100}{50} = 2\text{amp.}$

Q.4 Answer:

The charge on the capacitor is $Q = -LC\frac{d^2Q}{dt^2}$

Solution:

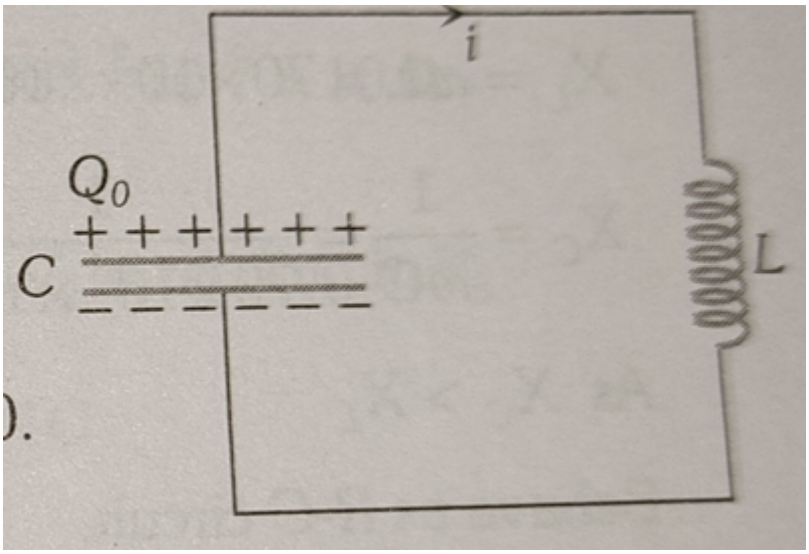
Since, $i = \frac{-dQ}{dt}$

$$\frac{Q}{C} - L\frac{di}{dt} = 0$$

$$\Rightarrow \quad \frac{d^2Q}{dt^2} + \frac{Q}{LC} = 0$$

$$\Rightarrow \quad Q = -LC\frac{d^2Q}{dt^2}$$

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Q.5 Answer:

$\frac{\varepsilon}{R}$

Solution:

Conserving flux in the inductor of inductance L

$L \cdot \frac{\varepsilon}{R} = Li'$

$\Rightarrow$ **New current after the switch is shifted to 2 is $\frac{\varepsilon}{R}$ which is the steady current in the circuit. So, current will not change.**

Q.6 Answer:

$50\text{ohm}, 4$ ampere.

Solution:

Here $\mathrm{X_L} = \omega \mathrm{L} = 2\pi \mathrm{fL} = 2\pi \times 50 \times \frac{0.4}{\pi} = 40\Omega; \mathrm{R} = 30\Omega$

$$\therefore \quad Z = \sqrt{R^2 + X_L^2} = \sqrt{30^2 + 40^2} = 50\Omega$$
$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{200}{50} = 4 \text{ A}$$

Q.7 Answer:

6.68×10^{-5} F

Solution:

$\because$ **Power P = VI** $\cos\phi = \frac{\mathrm{V}^2}{\mathrm{Z}} \cos\phi$

$$\therefore \quad Z = \frac{V^2 \cos\phi}{P} = \frac{(220)^2(0.5)}{(440)} = 55\Omega$$

Also power factor, $\cos\phi = \frac{\mathrm{R}}{\mathrm{Z}} \Rightarrow \mathrm{R} = \mathrm{Z}\cos\phi = (55)(0.5) = 27.5\Omega \therefore \quad X_L = \sqrt{Z^2 - R^2} = \sqrt{(55)^2 - \left(\frac{55}{2}\right)^2} = \frac{55\sqrt{3}}{2}\Omega$ **when power factor is** 1.0 **then** $\mathrm{X_L} = \mathrm{X_C}$

so $\mathrm{X_L} = \frac{1}{\omega \mathrm{C}} \Rightarrow \mathrm{C} = \frac{1}{\omega \mathrm{X_L}} = \frac{1}{(314)\left(\frac{55\sqrt{3}}{2}\right)} = 6.68 \times 10^{-5}$ F

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Q.8 Answer:

$\sqrt{481}$ V

Solution:

$V_L = 36$ V

Voltmeter between A and C

$$\sqrt{V_L^2 + V_R^2} = 39$$

Voltmeter between B and D

$$\sqrt{V_C^2 + V_R^2} = 25$$

From eq. (i) and (ii)

$$V_R^2 = 39^2 - 36^2 \Rightarrow V_R = 15 \text{ V}$$

From eq. (iii) and (iv)

$$V_C^2 = 25^2 - 15^2 \qquad \Rightarrow V_C = 20 \text{ V}$$
$$V_{\text{rms}} = \sqrt{(V_L - V_C)^2 + V_R^2} = \sqrt{16^2 + 15^2} = \sqrt{481} \text{ V}$$

Q.9 Answer:

Potential difference across R is 80 V

Solution:

Potential difference across the resistance $\mathrm{V_R = iR = 5 \times 16 = 80}$ **Volt**

Potential difference across the inductor $\mathrm{V_L = i \times (\omega L) = 5 \times 24 = 120}$ **Volt**

Potential difference across the capacitor $\mathrm{V_C = i \times (1/\omega C) = 5 \times 12 = 60 Volt}$

Q.10 Answer:

$V_{XY}^{rms} = V_0\sqrt{\frac{3}{2}}$

Solution:

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$$(\text{ac})\ V_{XY} = V_X - V_Y = (V_{XY})_0 \sin(\omega + \phi_1)$$

$$(V_{XY})_0 = \sqrt{V_0^2 + V_0^2 - 2V_0^2 \cos\frac{2\pi}{3}} = \sqrt{3}V_0$$

$$(V_{XY})_{rms} = \frac{(V_{XY})_0}{\sqrt{2}} = \sqrt{\frac{3}{2}}V_0$$

$$V_{YZ} = V_Y - V_Z = (V_{YZ})_0 \sin(\omega t + \phi_2)$$

$$(V_{YZ})_0 = \sqrt{V_0^2 + V_0^2 - 2V_0^2 \cos\frac{2\pi}{3}} = \sqrt{3}V_0$$

$$(V_{YZ})_{rms} = \frac{(V_{YZ})_0}{\sqrt{2}} = \sqrt{\frac{3}{2}}V_0.$$

Q.11 Answer:

both circuits must be having same value of capacitance and inductor

Solution:

$\mathrm{X_L} = \mathrm{X_C}$ **at resonance**
$\therefore \frac{\mathrm{X_L}}{\mathrm{X_C}} = 1$ **for both circuits.**

Q.12 Answer:

resistor

Solution:

X **is resistor and** Y **is a capacitor. Since the current in the two devices is the same (0.5A at 220 volt) When** R **and** C **are in series across the same voltage then** $R = X_C = \frac{220}{0.5} = 440\Omega$

$$\Rightarrow \mathrm{I_{rms}} = \frac{\mathrm{V_{rms}}}{\sqrt{\mathrm{R^2 + X_C^2}}} = \frac{200}{\sqrt{(440)^2 + (440)^2}} = \frac{220}{440\sqrt{2}} = 0.35\ \mathrm{A}$$

Q.13 Answer:

capacitor.

Solution:

X **is resistor and** Y **is a capacitor. Since the current in the two devices is the same (0.5A at 220 volt) When** R **and** C **are in series across the same voltage then** $R = X_C = \frac{220}{0.5} = 440\Omega$.

Q.14 Answer:

$0.35\ \mathrm{A}$

Solution:

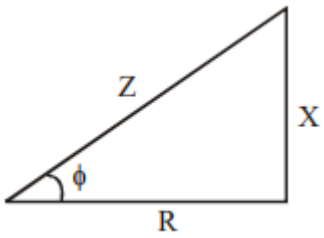
X **is resistor and** Y **is a capacitor. Since the current in the two devices is the same (0.5A at 220 volt) When** R **and** C **are in series across the same voltage then** $R = X_C = \frac{220}{0.5} = 440\Omega$.

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Q.15 Answer:

500Ω

Solution:



Impedance, $Z = \frac{V_0}{I_0} = \frac{100}{100\times10^{-3}} = 1000\Omega$

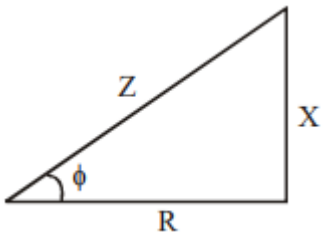
$$R = Z\cos\phi = 1000\cos\left(\frac{\pi}{3}\right) = 500\text{ohm}$$
$$X = Z\sin\phi = 1000\sin\left(\frac{\pi}{3}\right) = 500\sqrt{3}\text{ohm}$$

$$P_{av} = V_{rms}I_{rms}\cos\phi = \frac{100}{\sqrt{2}} \times \frac{100}{\sqrt{2}\times 1000} \times \cos\left(\frac{\pi}{3}\right) = 5 \times \frac{1}{2} = 2.5\text{watts}$$

Q.16 Answer:

$500\sqrt{3}\Omega$

Solution:



Impedance, $Z = \frac{V_0}{I_0} = \frac{100}{100\times10^{-3}} = 1000\Omega$

$$R = Z\cos\phi = 1000\cos\left(\frac{\pi}{3}\right) = 500\text{ohm}$$
$$X = Z\sin\phi = 1000\sin\left(\frac{\pi}{3}\right) = 500\sqrt{3}\text{ohm}$$

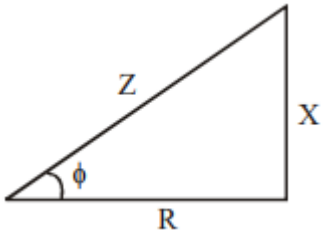
$$P_{av} = V_{rms}I_{rms}\cos\phi = \frac{100}{\sqrt{2}} \times \frac{100}{\sqrt{2}\times 1000} \times \cos\left(\frac{\pi}{3}\right) = 5 \times \frac{1}{2} = 2.5\text{watts}$$

Q.17 Answer:

2.5 W

Solution:

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Impedance, $Z = \frac{V_0}{I_0} = \frac{100}{100 \times 10^{-3}} = 1000\Omega$

$$R = Z\cos\phi = 1000\cos\left(\frac{\pi}{3}\right) = 500\text{ohm}$$
$$X = Z\sin\phi = 1000\sin\left(\frac{\pi}{3}\right) = 500\sqrt{3}\text{ohm}$$

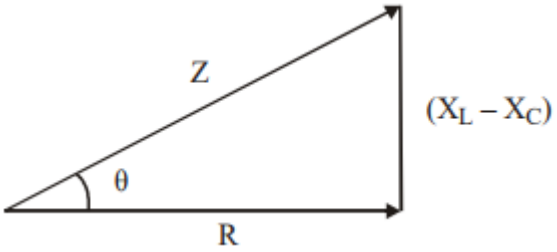
$$P_{av} = V_{rms}I_{rms}\cos\phi = \frac{100}{\sqrt{2}} \times \frac{100}{\sqrt{2} \times 1000} \times \cos\left(\frac{\pi}{3}\right) = 5 \times \frac{1}{2} = 2.5\text{watts}$$

Q.18 Answer:

$(A) - s, t (B) - p, r, s, t (C) - q, s (D) - p, r, s, t$

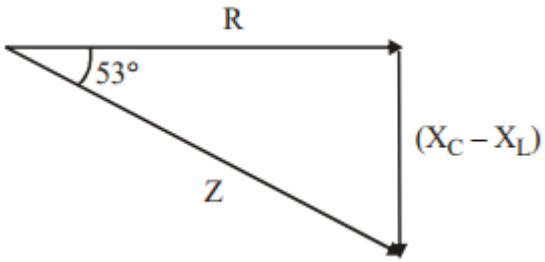
Solution:

$(A) - st, (B) - prst, (C) - qs, (D) - prst$



$|V_L - V_C| = 60 \Rightarrow V_R = \sqrt{100^2 - (60)^2} = 80\text{ V}$
$\frac{V_R}{V} = \cos\phi = 0.8 \Rightarrow \phi = 37^\circ, z = \frac{5}{4}R$

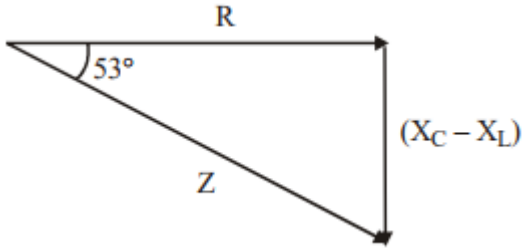
No other conclusions possible.



(b) $V_R = 60 \Rightarrow |V_L - V_C| = \sqrt{V^2 - V_R^2} = 80,\ V_L = 60 \Rightarrow V_C = 140 \Rightarrow$ **capacitative** $\Rightarrow$ **current leads,** $\omega < \omega_0 \Rightarrow \cos\phi = \frac{V_R}{V} = 0.6; \phi = 53^\circ, z = \frac{5}{3}R$

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(C) $V_L = V_C \Rightarrow$ **Resonance**



(D) $(V_L - V_C)^2 = V^2 - V_R^2 = 100^2 - 60^2$

$V_L - V_C = \pm 80; \cos\phi = \frac{60}{100} = 0.6; \phi = 53^\circ$

$V_C > V_L$ (given): Capacitive, current leaves $\Rightarrow \omega < \omega_0$

$\frac{R}{Z} = 0.6 \Rightarrow Z = \frac{R}{0.6} = \frac{5}{3}R$

Q.19 Answer:

—

Solution:

Here, height of water, $h = 300$ m

Rate of volume flow of water, $\frac{dV}{dt} = 100\ \mathrm{m^3/s}$

Turbine-generator efficiency, $\eta = 60\% = 0.6$

PE of water of mass m **at height** h**, i.e.,** $W = mgh$

Thus, hydroelectric power,

$$P_h = \frac{dW}{dt} = \frac{d}{dt}(mgh)$$

$$= \frac{d}{dt}(V\rho gh)$$

$$(\text{as } m = V\rho\ )$$

$$= \rho gh \times \frac{dV}{dt}$$

$$\left(1000\ \mathrm{kg/m^3}\right)\left(9.8\ \mathrm{m/s^2}\right)(300\ \mathrm{m})\left(100\ \mathrm{m^3/s}\right)$$

$$= 29.4 \times 10^7\ \mathrm{W}$$

$$\text{Electric power } = \eta \times P_h = 0.6 \times \left(29.4 \times 10^7\ \mathrm{W}\right)$$

$$= 176.4 \times 10^6\ \mathrm{W} = 176.4\mathrm{MW}.$$

Q.20 Answer:

—

Solution:

In an ideal transformer $E_S I_S = E_p I_p$

But $I_s = \frac{E_s}{Z_s}$ **and** $I_p = \frac{E_p}{Z_p}$

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$\therefore \quad E_s \cdot \frac{E_s}{Z_s} = E_p \cdot \frac{E_p}{Z_p} \quad \therefore \frac{E_s}{E_p} = \sqrt{\frac{Z_s}{Z_p}} = \sqrt{\frac{100}{10000}} = \frac{1}{100} = 0.1$

$\therefore \quad \frac{E_S}{E_p} = 0.1 = 1 \times 10^{-1}$. **That is** $\frac{E_s}{E_p} = \frac{N_s}{N_p} = 1 \times 10^{-1}$

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